

Crime Scene

Puh-Puh-Puh P. DIDDY!

As a freelance investigator, you're often taking on odd jobs, jumping around from place to place. However, after a relatively uneventful 2024, you've begun to settle down in the great city of Los Angeles. Between the sun-soaked coastline of beaches and glittering entertainment industry, what isn't there to enjoy in Los Angeles?

Two words. Baby oil.

It's in our food. It powers our stoves. It fills our bathtubs. The whole city runs on baby oil. Yet, recently, baby oil has been nowhere to be seen. The baby oil magnate — Sean "P. Diddy" Combs — has missed his yearly baby oil worship for the first time since he was born in 1969. Word is, he hasn't been spotted outside his house for the past four months.

Something's wrong. You decide to investigate the disappearance of the city's number one baby oil supplier.

Scoping around his mansion, you initially notice zero signs of disturbances. Then, you see it. The infamous *Diddy Window* on the second floor — shattered in a hundred pieces. Your stomach drops. *WHO WOULD DO SUCH A THING?!???*

Quickly, you regather yourself. You must find the baby oil. For the city of Los Angeles.

Fashioning a baby-oil-infused grapple hook, you enter the second floor of Diddy's house. Once again, you sense no disturbances. Descending to the first floor, you pick up 12 powders, but no sign of Diddy or his baby oil. Investigating further, you enter the basement of his house. There, you discover 5 plastics, 4, fibers, 4 hairs — but no baby oil.

You're about to give up, but at that moment, a painting catches your eye: *The Last Supper*. You'd only heard about it in folklore, yet here it was, ready for you to behold your eyes upon it. However, something isn't right. The P. Diddy you knew never wore his watch on his right arm, as it would get in the way of his baby oil application. Instantly, you tear down the poorly made replica, and your jaw drops.

In front of you. P. Diddy. Lies. Dead. You fall to your knees. The man who provided for the city of Los Angeles, killed at the hands of some unknown suspect. In his hands lay a note: "*it's mine now*"

You're filled with an uncontrollable rage. Yet, the killer could have escaped through any of Diddy's baby oil delivery tunnels expanding throughout the whole city from his house.

No matter. You're determined to bring the killer to justice.

After reading every last one of Diddy's journals and autobiographies, you've narrowed your search down to five suspects. Listed below is all the information you could find about each of them.

Do it for the baby oil.

.5 Dolla

Curtis James Jackson III, commonly known by his stage name “.5 Dolla,” was born and lives in Queens, New York. However, as a renowned CD producer, he often finds himself traveling to LA to record his albums. A borderline narcissist, he’s never seen without his “Curtis” album in his Grammy-engraved CD case. An avid lover of nature, he enjoys spending most of his time outside, tending to horses on his ranch in Connecticut and wearing cotton jumpers. However, he loves one thing more than nature: money. A lyrical genius, he’s a follower of bedtime journaling. His latest journal entry might be his most revolting yet; “Get rich or die tryin’.” Ever since 2007, he’s had a poor relationship with Diddy, often challenging him to Plaster of Paris sculpt offs. However, Diddy’s always gotten the best of him, and he’s begun to train to challenge Diddy in a new subject: competitive gardening. Although, .5 Dolla’s never shied away from shady business. Ever since the age of 11, he’s been training in street fighting, and when he wasn’t fighting in the ring, he was selling crack on the strip.

Fingerprint: radial loop

Blood Type: B-

Pen: 1

Suspect Statement: “I ain’t do it!”

Jeffrey

Jeffrey was also born in New York City, however, unlike .5 Dolla, he tends to stay out of the spotlight. An incredibly intellectual individual, he’s been set for life ever since a young age. He’s very artistic, having played piano and violin his whole life, and possessing a knack for developing homemade photographs. However, because of this, he’s started to develop arthritis as he pushes 70 years of age, requiring daily arthritis medication. As his day job, he used to work as a teacher but soon outgrew his old job to open his own consulting firm. He’s still in love with his professor-esque look, however, constantly wearing denim jeans and using Hagaromo Chalk to formulate his master plan on blackboards. His master plan you may ask? Well, no one really knows that for sure. He’s often taking leave off for days at a time claiming he needs to pick up his daily diuretic pills, but he’s seen using the bathroom several times a day? No one really knows for sure what he’s doing. But his private jet tracker shows he’s traveling to the Caribbean, and he often comes back with ointment to treat his bug bites.

Fingerprint: tented arch

Blood Type: O-

Pen: 2

Suspect Statement: “What was I doing in LA? Well wouldn’t you like to know. I’ll tell you if you pay me \$10,000.”

Aneesh Iyer

Aneesh is a senior at Mason High School. To be honest, he probably didn't commit the crime, but he's so annoying that he's definitely the type of menace to deprive the city of Los Angeles of its precious baby oil commodity. An avid member of the no-sleep movement, he's always seen drinking coffee with a styrofoam cup in hand. With all his extra time, he's committed to becoming a mad scientist, sporting a polyester lab coat and lab goggles. He's even managed to find himself a pet bat.... somehow. In the lab, he claims he's cooking up an experiment for ISEF, but he's just reacting baking soda with vinegar. However, he's recently discovered an experiment requiring a copious amount of baby oil — an amount that doesn't exist in Ohio. For the past three years, he's boasted about his perfect attendance, but this year, he's taken a week off to visit campuses in California.

Fingerprint: double loop whorl

Blood Type: A+

Pen: 3

Suspect Statement: "Sure (when asked if he committed the crime)."

Kardash Kim

Kardash is a media celebrity on international television status. Born and raised in Los Angeles, she's one of the few people to live in LA their whole life and find worldwide success. She's recently come down with a bad cold, but as an actor, she often has to prepare for impromptu shoots, leading to her taking excessive amounts of cough medicine. She's also a model, always making sure to have acrylic nails applied and her favorite prop, interestingly, is a grocery bag. Of course, she has to pay the bills as well, owning a \$60 million dollar mansion, so she's recently taken up a deal with Bath and Body Works to advertise detergent and body wash. Despite this she's missing just one thing money can't buy: baby oil. There's only one way to get it, and due to the beef between her husband — Yekhan East — and Diddy, she has no way of acquiring the coveted item.

Fingerprint: plain whorl

Blood Type: O+

Pen: 4

Suspect Statement: "I was busy shooting Keeping up with the Kims. There's no way I would have time to break into his house."

Lebron James Jr.

Lebron Raymone "Bronny" James Jr. is a professional basketball player for the Los Angeles Lakers. Unfortunately, he's not doing too hot, and it doesn't help that his father is Lebron James. Because of all the stress, he's begun to develop bipolar disorder, having to skip practice to take his medication. Additionally, he also sprained his finger, and he can be seen walking around with an ice pack on his finger. He's gotten really desperate to get playing time; when drinking red bull didn't work, he was at a loss for what to do. Word on the street is he's resorted to the black market, having been spotted carrying around ziploc bags of an unknown substance. For a two week period, he was "out there hooping like his papa," but since then, he hasn't been spotted carrying a ziploc bag, and declined back to his former benchwarmer status.

Fingerprint: ulnar loop

Blood Type: AB-

Pen: 5

Suspect Statement: "I caught my teammate sleeping with my mama, and I knew I had to ball out. But then I realized...."

Section A: Powders [85 pts]

1. Identify the following powders found at the crime scene by name and chemical formula; then, indicate which suspect(s) the sample incriminates. [55]

Sample	Solubility	pH	HCl	NaOH	Iodine	Benedict's	Flame Test	Observations
Powder 1			Yes				Bright glow	
Powder 2	Yes			Clear		Bubbles in heat		
Powder 3							Red	
Powder 4	No		NR	Blurry		Blue		Lumpy
Powder 5	Yes	6			Yellow		Wisp when removed from flame	
Powder 6						Blue	Caramelized	Sweet smell
Powder 7	Yes		Yes		Clears up			
Powder 8		8	NR	Clear		Blue	Yellow	
Powder 9		7			Yellow			Cubic crystals
Powder 10							Green	
Powder 11	No			Blurry				Crystals

Sample	Identity [3]	Formula [1]	Suspect(s) Incriminated [1]
A1	Calcium carbonate	CaCO_3	Jeffrey
A2	Sodium bicarbonate	NaHCO_3	Aneesh
A3	Lithium chloride	LiCl	Bronny, Jeffrey
A4	Calcium sulfate	CaSO_4	.5 Dolla
A5	Ammonium chloride	NH_4Cl	.5 Dolla, Jeffrey, Kardash
A6	Sucrose	$\text{C}_{12}\text{H}_{22}\text{O}_{11}$	Bronny
A7	Sodium carbonate	Na_2CO_3	Kardash
A8	Sodium acetate	$\text{NaC}_2\text{H}_3\text{O}_2$	Jeffrey
A9	Sodium chloride	NaCl	None
A10	Boric acid	H_3BO_3	Jeffrey
A11	Calcium nitrate	$\text{Ca}(\text{NO}_3)_2$	Bronny

2. Magnesium sulfate is commonly known as epsom salts, but that isn't entirely true. Write out the entire chemical formula for epsom salts. [2]



3. Describe the use of cobalt blue glass when conducting a flame test. [2]

It filters out yellow light, allowing other colors from other metals or ions to be seen in the flame. This can prevent contamination or amplify ions of interest in the flame.

4. Write out the balanced equations for the following reactions [4]

- a. Calcium sulfate and hydrochloric acid



- b. Sodium acetate and water



5. Rank the following from most moles to least: 4 grams of sodium acetate, 10 grams of cornstarch, 5 grams of hydrated calcium sulfate, 1 gram of boric acid. [4]

Cornstarch, sodium acetate, hydrated calcium sulfate, boric acid

6. Identify if the following mixtures are soluble or insoluble in water, and if it is insoluble, what compound causes it to be insoluble. [6]

- a. Sodium acetate and calcium sulfate [2]

Insoluble, calcium sulfate since they will not react

- b. Sodium chloride and calcium nitrate [2]

Soluble

- c. Ammonium chloride and calcium nitrate [2]

Soluble

7. Write out the reaction that occurs between benedict's solution and a reducing sugar. [2]



8. List the chemical formula for each of the following sugars. [3]

- a. Glucose [1]



- b. Galactose [1]



- c. Fructose [1]



9. What powder is produced in the Solvay process? Identify the steps/reactions of the process. [4]

Sodium Carbonate [1]



10. Give the molecular geometry for the following powders. [3]

- a. Boric acid

Trigonal planar

- b. Sodium bicarbonate

Trigonal planar

- c. Lithium chloride

Linear

Section B: Plastics [33 pts]

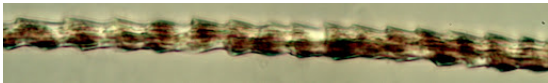
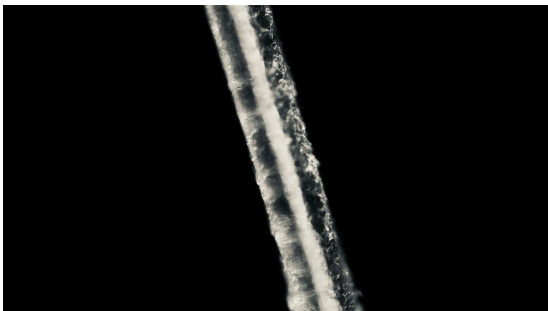
11. Based on the float/sink test results in various solutions (veg. oil ($d = 0.92 \text{ g/mL}$), 46% IPA ($d = 0.94 \text{ g/mL}$), water ($d = 1.00 \text{ g/mL}$), and 10% NaCl ($d = 1.07 \text{ g/mL}$)), identify the following plastics found at the crime scene by name and recycling code. Then, indicate which suspect(s) the sample incriminates. [15]

Sample	Veg. Oil	46% IPA	Water	10% NaCl	Observations	Identity [3]	Code [1]	Suspect(s) Incriminated [1]
B1	Sink	Float				LDPE	4	Bronny, Kardash
B2				Sink	Dissolves in acetone	PVC	3	Jeffrey
B3				Sink	Self extinguishes in flame	PC	7	.5 Dolla, Aneesh

12. What is the process of vulcanization? [2] adding sulfur to raw rubber [0.5] and heating it [0.5] to form cross-links between rubber molecules [0.5] and strengthen/harden rubber [0.5].
13. Cross-linking transforms polymers into [soluble/insoluble] and [linear/interconnected] structures. [2]
14. [T/F] Thermoplastics can only be heated & shaped once. [1]
15. Order the following plastics in order of highest to lowest tensile strength: PS, PP, PVC, PC. [2] PC, PVC, PS, PP
16. PP and PETE are slurry polymers. What is slurry polymerization? [3] the polymerization of a monomer in a liquid medium, where the resulting polymer forms as solid particles suspended in the liquid phase.
17. The copper wire flame test can be used to detect the presence of _____ in materials like PVC. [2] Chlorine
18. PETE used to be made from the monomers dimethyl terephthalate and ethylene glycol. Now, instead of dimethyl terephthalate, what monomer is commonly used? [2] What is the key difference between the two? [2] terephthalic acid [2] when using DMT, transesterification is used in the polymerization process.
19. PC is synthesized via [addition/condensation] polymerization. [1] What are its monomer units made of? [1] propylene

Section D: Hairs [36 pts]

28. Identify the following fibers at the crime scene by name; then, indicate which suspect(s) the sample incriminates. [12]

Sample	
D1	
D2	
D3	

Sample	Identity [3]	Suspect(s) Incriminated [1]
D1	Bat	Aneesh
D2	Human	None [1], or Kardash [3] (because she bleached her hair white)
D3	Horse	Jeffrey, .5 Dolla

29. How can you tell whether a hair has been bleached or not based on cuticle? [2] cuticle will appear rough and lifted under microscope.
30. [T/F] Analyzing root under microscope can determine whether hair was ripped & if there was struggle. [1]

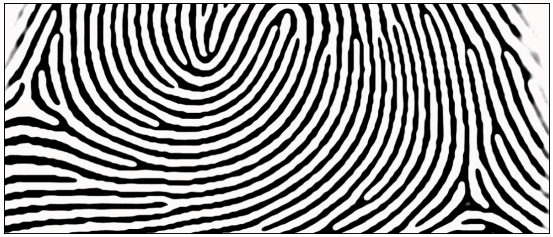
31. [cow/squirrel/horse/bat] hairs are used for fishing lines. [1]
32. Identify the four stages of the hair growth cycle in order and the percentage (or amount if applicable) of hairs in each stage. [5] anagen [0.5] 85% [0.5], catagen [0.5] 2-4% [0.5], telogen [0.5] 10-15% [0.5], exogen [0.5] 50-100 hairs per day [0.5] +1 for right order total 5 pts
33. Pheomelanin is the pigment responsible for _____-colored hair, eumelanin is the pigment responsible for _____-colored hair. [2] red [1], dark [1]
34. What is glabrous skin? Which parts of the body have glabrous skin? [3] skin that lacks any hair [2]. Lips, soles of feet, palms of hands [1]
35. Gray hair results from a lack of _____. [1] tyrosinase

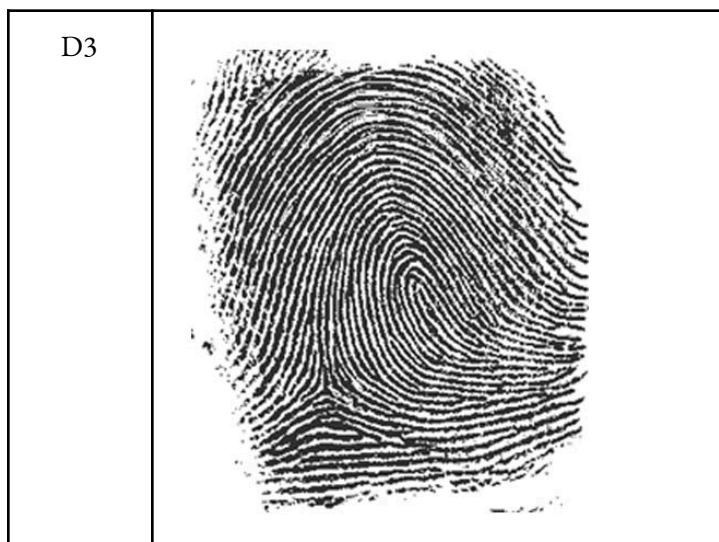


36. What disease/phenomenon is shown in the above picture? [2] postmortem hair banding
37. Why does burning hair smell? [3] high sulfur content released from keratin proteins breaking down in heat
38. Identify what animals best fit with the medullary indexes listed below. [4]
- a. 0.33 human
 - b. 0.58 (fragmented medulla) cat
 - c. 0.79 (lattice medulla) rabbit
 - d. 0.85 (cellular medulla) deer

Section I: Fingerprints [39 pts]

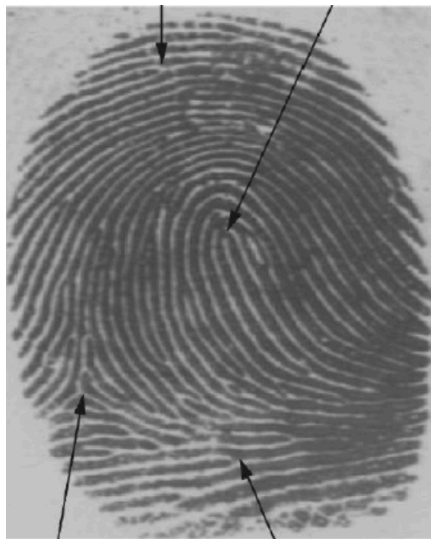
68. Identify samples I1-I3 using the provided materials. All of the prints are from the suspect's right hand. Then, indicate which suspect(s) the sample incriminates. [12]

Sample	
D1	
D2	



Sample	Identity [3]	Suspect(s) Incriminated [1]
D1	Plain arch	None
D2	Double loop whorl	Aneesh Iyer
D3	ulnar loop	Lebron James Jr.

69. What is the scientific name for the study of fingerprints? [2] dermatoglyphics.
70. Someone's left hand has a whorl, loop, arch, arch, and whorl, and their right hand has a arch, arch, whorl, loop, and loop (in order from pinky to thumb). Using the Henry Classification System, what is their primary fingerprint classification? [4] 18
71. For each of the following diseases, describe the etiology and some of its clinical manifestations. [6]
- adermatoglyphia: the absence of ridges on fingers, toes, palms & sole [1]. Caused by mutations in SMARCAD1 [1]
 - Naegeli Syndrome: award points for listing any of the following symptoms: results in absent fingerprints, thickening of palms & soles, decreased sweating, heat intolerance, hyperpigmentation, brittle nails, abnormally colored teeth, & early tooth loss [1]. Caused by mutations in KRT14 gene. [1]
 - Dermatopathia Pigmentosa Reticularis: results in reticulate hyperpigmentation, noncicatricial alopecia, & onychodystrophy, and complete absence of fingerprints [1]. Caused by mutations in KRT14 gene. [1]



72. Identify each of the minutiae in the image above from left to right, top to bottom. [4] A: ridge ending [1], B: core [1], C: delta [1], D: Ridge bifurcation [1]
73. [T/F]: Loops are the most dominant in Asians. [1]
74. [T/F]: Similarities exist between parents and child's fingerprints. [1]
75. [T/F]: Animal fingerprints like koala prints are extremely similar to human fingerprints. [1]
76. Describe the life cycle of fingerprints (a.k.a when and what happens at each stage of formation) [4] start developing 10th week in womb [1]. Ridges form at month 6 [1] from touching surrounding structures [1]. Fully formed between 19-23rd week [1]
77. Name the fingerprint development method best used in each of the following scenarios. [4]
- Fresh prints found on cardboard iodine fuming [1]
 - Old prints found a glass table cyanoacrylate fuming [1]
 - Dry print found on clothing ninhydrin [1]
 - Prints found on smooth, wet plastic small particle reagent [1]

Section K: Analysis [148 pts]

84. Who committed the crime? Be sure to mention who is guilty and who is innocent, all evidence discovered at the crime scene and why they tie back to certain suspects, potential motives, and any other reasoning that led to your conclusion. Feel free to address suspects by first name.

.5 Dolla is innocent [5]

- PC from CD cases
- Horse hair
- Cotton from jumpers
- Calcium sulfate from Plaster of Paris
- Ammonium chloride from gardening
- Cocaine from mass spectrometry

Jeffrey is guilty [20]

- Horse hair from violin
- Sodium acetate from photograph developing
- Lithium chloride from arthritis medication
- Calcium Carbonate from chalk
- PVC from blackboard
- Ammonium chloride from diuretics
- PP from pill bottle
- Boric acid from bug bite ointment
- Cotton from denim jeans
- Pen matches pen found at crime scene

Aneesh Iyer is Innocent [5]

- PS from styrofoam
- Polyester from lab coat
- PC from goggles
- Bat hair
- Sodium bicarbonate from baking soda
- Double loop whorl found at crime scene (+3 for mentioning that could be that Diddy had many other people at his house)

Kardash Kim is innocent [5]

- Ammonium chloride from cough medicine
- PMMA from acrylic nails
- LDPE from grocery bag
- HDPE from detergent bottle
- Sodium carbonate from body wash
- O+ blood at crime scene (+2 for mentioning that O+ blood type is very common or could be Diddy's blood)
- Blonde hair found at crime scene

Lebron James Jr. is innocent [5]

- Lithium chloride from bipolar
- Calcium nitrate from ice pack
- Sucrose from red bull
- LDPE from ziploc bags
- Ulnar loop found at crime scene (+3 for mentioning that could be that Diddy had many other people at his house)

+2 per evidence mentioned (no double jeopardy; if a sample was misidentified and used in the analysis, it earns points)

+1 per connection to suspect (ex. MgSO_4 because uses Epsom salts)

+1 per rationale for why certain evidence is not incriminating (ex. Margo Pay cut her leg, hence blood type matches crime scene)

+3 per correct statement regarding motive, alibi, or lack thereof, etc.